MEDIUM ALGEBRA

Algebra

06

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ALGEBRA

A total of 2 squares each have side length r. A total of 6 equilateral triangles each have side length t. None of these squares and triangles shares a side. The sum of the perimeters of all these squares and triangles is 210. Which equation represents this situation?

A. $6r + 24t = 210$

B. $2r + 6t = 210$

C. $8r + 18t = 210$

D. $6r + 2t = 210$

Q2 ALGEBRA

The cost to rent a bus from Company X is $\$950$ for the first 3 hours and an additional $\$50$ per hour for each hour after the first 3 hours. If the total cost to rent the bus from Company X for t hours, where $t > 3$, is $\$1{,}150$, which equation represents this situation?

A. $950(t - 3) + 50t = 1{,}150$

B. $950(3t) + 50t = 1{,}150$

C. $950 + 50(t - 3) = 1{,}150$

D. $950 + 50(3t) = 1{,}150$

Q3 ALGEBRA

In the xy-plane, the graph of the linear function f contains the points $0, 2$ and $8, 34$. Which equation defines f, where $y = fx$?

A. $fx = 2x + 42$

B. $fx = 32x + 36$

C. $fx = 4x + 2$

D. $fx = 8x + 2$

Q4 ALGEBRA

$$y = 2x - 3$$
$$3y = 5x$$

In the solution to the system of equations above, what is the value of y ?

A. -15

B. -9

C. 9

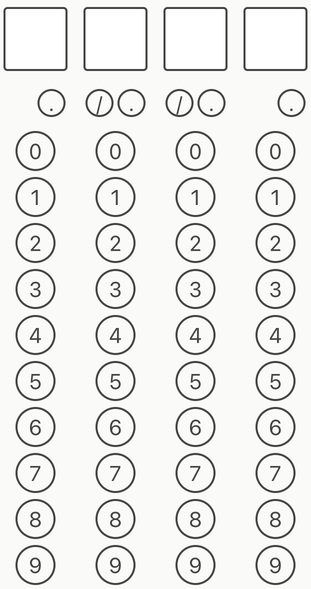
D. 15

Q5

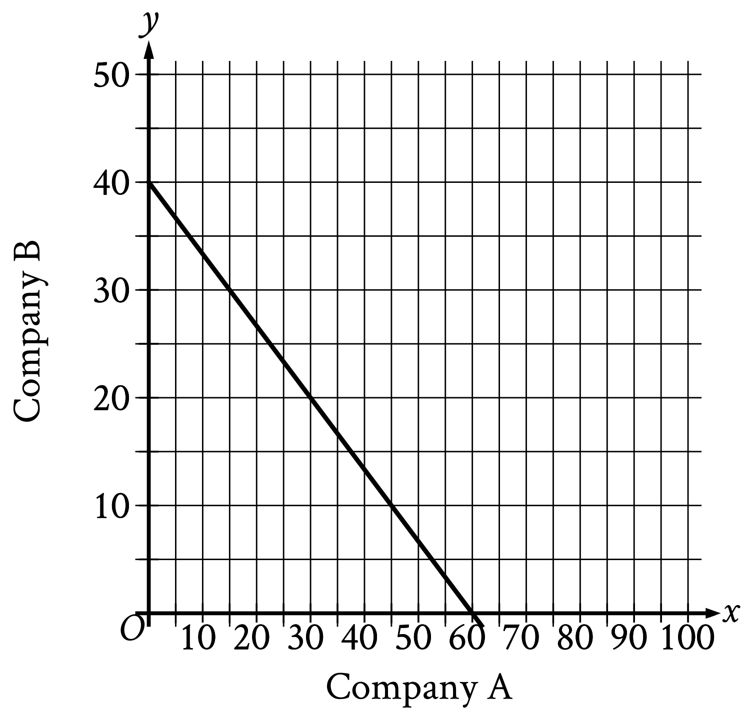
GRID-IN ALGEBRA

An event planner is planning a party. It costs the event planner a onetime fee of $\$35$ to rent the venue and $\$10.25$ per attendee. The event planner has a budget of $\$300$. What is the greatest number of attendees possible without exceeding the budget?

STUDENT-PRODUCED RESPONSE



Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- The line slants down from left to right.
- The line passes through the following points:
  - (0 comma 40)
  - (60 comma 0)

The graph shows the relationship between the number of shares of stock from Company A, x, and the number of shares of stock from Company B, y, that Simone can purchase. Which equation could represent this relationship?

A. $y = 8x + 12$

B. $8x + 12y = 480$

C. $y = 12x + 8$

D. $12x + 8y = 480$

Q7 ALGEBRA

If $4x - \frac{1}{2} = -5$, what is the value of $8x - 1$?

A. 2

B. $-\frac{9}{8}$

C. $-\frac{5}{2}$

D. -10

Q8

ALGEBRA

Population of Greenleaf, Idaho

Year	Population
2000	862
2010	846

The table above shows the population of Greenleaf, Idaho, for the years 2000 and 2010. If the relationship between population and year is linear, which of the following functions P models the population of Greenleaf t years after 2000?

A. $P(t) = 862 - 1.6t$

B. $P(t) = 862 - 16t$

C. $P(t) = 862 + 16(t - 2{,}000)$

D. $P(t) = 862 - 1.6(t - 2{,}000)$

Q9 ALGEBRA

$$y = 2x + 10$$

$$y = 2x - 1$$

At how many points do the graphs of the given equations intersect in the xy-plane?

A. Zero

B. Exactly one

C. Exactly two

D. Infinitely many

Q10 ALGEBRA

The length of a rectangle is 50 inches and the width is x inches. The perimeter is at most 210 inches. Which inequality represents this situation?

A. $2x + 100 \leq 210$

B. $2x + 100 \geq 210$

C. $2x + 50 \leq 210$

D. $2x + 50 \geq 210$